



Sigfox 0G technology and LoRaWan combined in Advanced (Water)Metering Infrastructures

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1. UnaBiz and the Convergence of IoT Technologies

IoT Convergence is about *‘making IoT Ecosystems interoperable to remove frictions and complexity for end users to drive Massive IoT adoption’*.

UnaBiz is positioned as a Massive IoT integrator capable of offering a single IoT Technology or combining multiple IoT Technologies (Sigfox OG, LoRaWAN, Cellular..) to support IoT Customers in achieving their primary objective: collecting IoT data at scale to make informed decisions and take actions.

Convergence aka the ‘capability to combine multiple IoT technologies’ aims to resolve three main business issues that IoT market is facing:

- No single IoT Technology is able to fully cover all IoT use cases
- No single IoT Technology has a pervasive coverage globally
- The integration of multiple IoT Technologies is complex

Based on feedback from both Utility customers and Service providers, we have observed that IoT Convergence can deliver significant value to the Water and Gas utility market segment.

2. Purpose

The purpose of this paper is to demonstrate the benefits of integrating LoRaWAN and Sigfox OG technology as complementary IoT technologies for connecting Advanced (Water) Metering Infrastructures (AMI) across both small and large territories.

Through this paper, we will demonstrate that the simultaneous deployment of Sigfox OG technology and LoRaWAN enhances the service performance level of AMI (index collection success rate) in a cost-effective way.

Additionally, this integration allows Water Meter manufacturers to upgrade their LoRaWAN Smart Water Meters with Sigfox OG technology, minimizing cost, effort and risk, through firmware upgrades, opening up new market opportunities and enhanced performance.

The assumption of this paper is that Sigfox OG technology and LoRaWAN are both available in the same territory. However, in some territories, only one of the two technologies is able to deliver the required coverage. **Each technology has proven to be capable** of being used as the **single option**.

3. Problems to be resolved

3.1 SLAs and Issues to Collect Data

3.1.1. Stringent Service Level Agreements (SLAs)

The deployment of Advanced Metering Infrastructure (AMI) is a high priority for utility companies globally, eliminating the need for manual meter reading, decreasing Non-Revenue Water, and increasing Customer satisfaction by leveraging near real time data for leakage detection and better consumption awareness.

As a consequence, some Utilities require more stringent Service Levels (hourly index collection SLA > 95 %) increasing the cost and complexity of IoT Network.

The key question to ask is 'how does one meet the higher SLA of IoT Networks that connect water meters at the massive scale while maintaining a positive ROI for the Utilities'.

3.1.2. Network Capacity

Utility networks are usually made up of a mixture of rural areas (low density of meters) where IoT network communication capacity is not an issue, and urban areas (high density of meters) where network communication capacity is limited (high gateway redundancy and risk of communication collisions).

EMEA regulation in the unlicensed band brings a limitation to Downlink Capacity, and full duplex communications (concurrent Uplinks and Downlinks) are not possible due to the allocated frequency agreement of International Telecommunications Union (I.T.U.).

3.1.3. Network Coverage

The radio coverage planning of a water metering deployment is particularly challenging as meters can sit outdoor, indoor or even deep indoor (such as underground). While LPWAN technologies are excellent at penetrating walls and buildings, a fraction of water meters may remain unconnected to the AMI due to radio performance limitations. Last but not least, no single LPWAN Technology has pervasive coverage globally.

3.1.4. Resistance to Interferences

It is not uncommon with radio technologies, for urban areas specifically, to experience what is known as “human made noise”. These noises, or interferences, can be local and temporary, or permanent and widespread in other areas.

This noise can be difficult to foresee and wireless technologies may not address interference issues with the same level of effectiveness.

3.1.5 Acknowledgment and Repetition of Data

As metering data can be missing due to capacity, coverage or interference issues, there is a need to build a solid data recovery scheme, based on message repetitions, leveraging time, frequency, and possibly technology diversity. It also relies on an advanced Network Server Adaptive Data Rate algorithm. Designing an efficient data recovery scheme is often a challenging task.

3.2 Engineering and logistics complexity (multi-investment and multi-SKU)

For meter manufacturers and/or solution providers, developing and maintaining products supporting several IoT technologies, requires engineering skills, investment and dedicated manpower. Leading to a variety of stock keeping units (SKUs).

As an example, a water meter product portfolio can be split into Wireless M-BUS, LoRaWAN, Cellular or Sigfox OG Network versions with different hardware and software implementations.

Investing in a single SKU multi-LPWAN radio communication meter, (Sigfox OG technology and LoRaWAN and possibly other LPWAN,), enables meter manufacturers to secure their long-term investment by accessing larger market segments with one product line. Adding Sigfox OG stack on top of LoRaWAN based devices does not require any hardware modification (it is a cost effective transition).

4. Recommendation: Sigfox OG and LoRaWAN collaboration to reach a higher service level performance

We do recommend the collaboration of Sigfox OG technology and LoRaWAN to resolve the issues discussed above:

- Network Coverage
- Network communication capacity
- Missing index recovery
- Interferences

It is not the goal of the paper to give a detailed recipe of the Sigfox + LoRaWAN optimized Network design. However, we are able to bring here the **key principles** of a fruitful and unique collaboration between the two technologies that can be replicated across Utility IoT Network deployments globally.

Network Coverage

Sigfox OG technology is a preferred choice to connect hard to reach meters in a cost-effective way. As a narrowband protocol at low data rate, Sigfox OG has the capability to connect remote water meters (long range) supporting high network communication capacity (4M+ Uplink messages per day per Gateway). Thus, a Sigfox-based solution requires a lower number of gateways to cover a target territory, as compared to other LPWAN Technologies. In most Regions, one at least of the two technologies is deployed with a wide coverage. This is an opportunity to rely on an existing network infrastructure (Sigfox or LoRaWAN) versus investing in a new Network roll-out (cost, delay, and complexity).

Network communication Capacity

LoRaWAN is an effective network technology in highly densified areas playing with spreading factors, time, frequency and gateway diversity. LoRaWAN's densification capability has a positive impact on device battery lifespan and communication capacity increase. Sigfox OG technology is also able to support high capacity Networks in any areas, with a higher battery consumption that can be compensated by a smart index collection scheme (small payload and low frequent communications)

Missing index recovery: both technologies can collaborate in the index recovery scheme to overcome duty cycle limitations on Downlink capacity and full duplex inability (a GTW cannot send and emit messages simultaneously). For instance, one technology can focus on Uplink transmissions sending daily indexes (and repetitions) where the other technology can focus on Downlink transmissions (requesting missing indexes at the end of a billing period).

Interference: rolling-out an hybrid LoRaWAN & Sigfox OG technology Advanced Metering Infrastructure concurrently utilizing narrowband (Sigfox) and wideband (LoRaWAN) technologies, increases the capability to overcome interference issues. This also increases the network gateway diversity and impacts the radio performance positively especially as Sigfox OG technology is highly resistant to interferences.

Power of the complementary technology set-up

Implementing Sigfox OG and LoRaWAN concurrently statistically increases the network performance and resilience, bringing a multi factor diversity

- Diversity in interference resistance (static and temporary interferers)
- Diversity in Gateways locations
- Diversity in Gateway redundancy
- Diversity in Frequency, time, and data rate
- Diversity in radio coverage performance
- Diversity in network availability

The detailed applicable network strategy will be **adapted to each utility project**: depending on the local radio conditions, SLA requirement, and **coverage availability** of each technology. Depending on the geographical location, different LPWAN technologies may not be available nationwide.

4.2 Example of Sigfox OG and LoRaWAN collaboration

Caveat: The below example is inspired from a **specific Business** case which does not apply as a general statement. The role split between Sigfox OG Technology and LoRaWAN **can be different depending on the local Business case**. We will discuss below an example where both technologies collaborate to optimize cost and performance. Again , each of the two technologies can be used as a **single option in any Region, with a different Network design leveraging each technology strengths**.

Whatever the Business case (single technology or collaboration), we are willing to demonstrate that embedding Sigfox OG technology and LoRaWAN in each Smart water meter has a business value bringing the flexibility to adapt to any Business situation.

In this example, LoRaWAN is actually used as the main source of data collection, relying on an hourly index that represents a significant amount of data, in compliance with several years device battery lifespan. In the target territory, a country wide Sigfox Network is already available.

An existing LoRaWAN Network is up and running. The customer is willing to increase the Level of service up to 99.8 % based on hourly indexes (SLA) while rolling out a new set of hundreds of thousands of thousands of water meters.

The key question is *‘how to upgrade (re-design) at best cost the actual IoT Network able to connect the new meters while reaching a higher SLA’*

In order to meet a higher SLA, the actual LoRaWAN Network needs to fix the below performance issues:

- Missing data in dense urban areas due to peaks of transmissions (collisions), and lack of gateway diversity in some areas
- Coverage issues for hard to reach clusters of scattered meters
- Presence of local temporary interferers

The assumption is that Water Meters embed Sigfox OG and LoRaWAN Technologies.

As said, the Sigfox OG Network is already deployed countrywide (it may have some local extensions).

Below are guidelines on how to combine both technologies in order to resolve the identified Network issues.

Recovery of missing hourly indexes: proposed scheme

- Each Water Meter sends LoRaWAN hourly indexes through Uplinks. For instance 4 Uplinks per day (each representing 24 hourly indexes with a 6 hours overlap between Uplinks).
- The application server checks the index completion, then requests through a Sigfox Downlink the missing hourly indexes of the month for the Water Meters involved.
- Then, the Water Meters involved send back the missing indexes through LoRaWAN Uplinks. The process of Sigfox recovering the missing hourly indexes can be set-up in the last days of each month.

Based on the above the LoRaWAN Technology is focused on Uplinks process sending indexes massively, where the Sigfox OG Network is specialized in asking for missing indexes in the last days of each month. The benefit is to preserve the LoRaWAN Gateway Uplink capacity by using the Sigfox OG Network for Downlinks.

Coverage

The Sigfox OG Network, which requires a limited number of gateways to cover a specific area, is designed to well cover the scattered clusters of meters which are at the limit of coverage. In that case, Sigfox-Only meters can be used.

- In remote and rural areas where Coverage is the main issue, Sigfox OG technology can be used for Uplink and Downlink transmissions.

- In this example, the Sigfox Network is available. Deploying a new LoRaWAN Network would not be as cost-effective.

The benefit of using the existing Sigfox OG Network is to reduce the overall cost of the Advanced Metering Infrastructure.

In summary, Sigfox OG technology is a perfect complement to LoRaWAN for the discussed Business Case

The dual mode set-up increases the index recovery SLA while optimizing the Advanced Metering Infrastructure cost as follows:

- Sigfox to recover missing hourly indexes in the last days of each month (Downlink requests only)
- Sigfox to be the primary technology at best cost (Uplink and Downlink) for hard to reach meters in rural and suburban areas
- LoRaWAN to be the primary the primary technology in very dense areas
- Combined two LPWAN technologies brings a multi factor diversity (site locations, radio performance, capacity, time, payload structure, frequency, GTW redundancy, Network redundancy), to overcome interference and network capacity/ available issues.
- It is much more cost effective to use an existing Sigfox Network in part of the territory than deploying a brand new LoRa Network to connect suburban and remote areas, given that increasing the SLA by a few % would dramatically increase the cost of LoRaWAN Network expansion.

4.3 Opportunity for Meter Manufacturers and Utility Companies

Based on the recent chipset platforms from Semtech and STMicroelectronics, it is now possible to have a dual-mode / dual-stack solution supporting LoRaWAN and the Sigfox OG technology hosted on one single hardware/BOM at no added cost, through a simple a firmware upgrade.

It is then up to the Utility customers to use one technology or the other dynamically at the same time, or not.

As utility companies are reluctant to be locked in with one technology, it is also an opportunity to offer a solution which can be switched over time (reversibility): either through a firmware upgrade or, preferably, dynamically and remotely in the field.

Having a dual-mode smart meter solution in the field derisks a smart metering infrastructure investment for all the stakeholders as a technology can be changed rapidly if it is no longer adapted to the needs, or becomes too expensive: business continuity is ensured.

Lastly, depending on the country, one of LPWAN Technologies may just not be available, or only delivers a poor limited radio coverage.

Thus, using available LPWAN Technologies versus deploying a new one from scratch is the best financial and technical option.

5. Next steps

5.1 Implementing LoRaWAN & Sigfox dual-mode

The implementation requires two steps:

- Upgrade existing LoRaWAN smart meter devices by adding the open-source Sigfox stack on top of the LoRaWAN stack
- Implement the communication and repetition scheme required by Utility customers onto the meter radio chipset firmware

5.2 Upgrade existing LoRaWAN meters with Sigfox stack

Implementation of the Sigfox stack requires the use of the below chipset references:

- Semtech LR1110, LR1120, LR1121, SX1261, SX1262
- ST Micro SoC (STM32WLE5 series)
- as well as any module using these chipsets (Murata, SJI...).

Additional requirements:

- Oscillator: TCXO or Quartz with a 20 ppm precision at operating temperature and including aging
- Memory size: from 1 to 8kB depending on the options

You will find the open source reference code for Semtech LRxx series at

[LRxx by Sigfox stack](#)

The integration with the Semtech **SX126x** series, requires the signature of a Non Disclosure Agreements with Semtech, giving access to the technical Application Note.

UnaBiz can connect Meter manufacturers to the right stakeholders at Semtech, STMicro and other module makers to implement the dual-mode Stack. Unabiz can also provide some support and sample code for the implementation of the Sigfox stack and its use.

5.2 Firmware strategy

Supporting LoRaWAN and Sigfox OG technology dual mode requires the implementation of a new embedded software strategy.

- Implement the communication and repetition scheme required by Utility customers onto the meter radio chipset firmware. In our example, collect efficiently missing data indexes at the end of the month for reach a demanding SLA.

6. Conclusion

In this paper, we learnt that:

- Utility customers may require a more stringent Level of Service that a single IoT Technology will not meet in any situations in a cost-effective way.
- The collaboration of Sigfox OG technology and LoRaWAN empowers IoT Service Providers to resolve Advanced Metering Infrastructure network performance issues (coverage, capacity, interference, index recovery) in a cost-effective way
- Global LPWAN coverage is fragmented depending on geography. The integration of LoRaWAN and Sigfox OG technology within a Smart Water Meter resolves the issue of inconsistent coverage availability. It is dramatically more cost effective to use the available LPWAN technology instead of investing in a brand new Network roll-out.
- Combining two LPWAN technologies brings a multi factor diversity (site locations, radio performance, capacity, time, payload structure, frequency, gateway redundancy, network redundancy), to overcome interference and

network capacity/ availability issues. It also de-risks strategically the investment decision by reducing drastically dependence on one technology only.

- The implementation of Sigfox and LoRaWAN dual mode is today the most straightforward and cost-effective process (via firmware upgrade) supported by the leading chipset vendors like Semtech and STMicroelectronics
- The integration of Sigfox OG technology and LoRaWAN dual-mode mitigates the investment risks for meter manufacturers in IoT, enabling them to access a larger market and provide enhanced coverage and performance to utilities companies across all regions.